

《数据库系统原理》实验报告（2）

题目：

学号	2452769	姓名	幸可函	日期	2026/3/31
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实验环境：

1. 操作系统：Windows 11
2. 容器平台：Docker Desktop for Windows（版本：4.65.0）
3. 数据库镜像：oceanbase/oceanbase-ce:latest
4. 数据库系统：OceanBase Community Edition 4.3.5.5

实验步骤及结果截图：

- 1、参照下表内容，在数据库中建立教学管理系统相关数据表。

表名：students		
字段	类型	约束
No	Integer	主键，非空
Name	Varchar	非空
Gender	Varchar	非空，只能为 Female 或者 Male
Age	Integer	非空
D_no	Integer	非空，外键

表名：depts		
字段	类型	约束
No	Integer	主键，非空
Name	Varchar	非空

表名：courses		
字段	类型	约束
No	Integer	主键，非空
Name	Varchar	非空
Credit	Integer	非空
D_no	Integer	非空，外键

表名：scores		
字段	类型	约束
S_no	Integer	非空，外键
C_no	Integer	非空，外键
Score	Integer	非空

注：该数据库及相关数据表在实验一中均有创建（但是表名均为该英文名词的单数形式），因此本次实验直接沿用，并没有重新创建一次。

但在实验一中，student 表要求 Gender 只能为 "F" 或 "M"，本次实验则是 "Female" 或 "Male"，故需重新修改表格：

```
obclient(root@sys)[teaching_management]> SELECT table_name, constraint_name, constraint_type FROM information_schema.TABLE_
CONSTRAINTS WHERE table_name="student";
+-----+-----+-----+
| table_name | constraint_name | constraint_type |
+-----+-----+-----+
| student   | PRIMARY        | PRIMARY KEY    |
| student   | student_OBCHECK_1774341836809447 | CHECK          |
| student   | student_OBFK_1774342734704192 | FOREIGN KEY    |
+-----+-----+-----+
3 rows in set (0.158 sec)

obclient(root@sys)[teaching_management]> alter table student drop constraint student_OBCHECK_1774341836809447;
Query OK, 0 rows affected (0.788 sec)

obclient(root@sys)[teaching_management]> alter table student modify column Gender varchar(6) check (Gender in ("Male","Fema
le"));
Query OK, 0 rows affected (1.468 sec)
```

- 2、使用比较运算符查询所有年龄在 21 岁以下的学生姓名及其年龄

```
obclient(root@sys)[teaching_management]> select Name, Age
-> from student
-> where Age < 21;
+-----+-----+
| Name   | Age   |
+-----+-----+
| Tommy  | 20    |
| Michael| 19    |
| Ale    | 18    |
| Billy  | 19    |
+-----+-----+
4 rows in set (0.011 sec)
```

3、查询选 2 号课程 (C_no='2') 且成绩在 80--90 的学号

```
obclient(root@sys)[teaching_management]> select S_no
-> from score
-> where C_no = '2' and Score between 80 and 90
-> ;
+-----+
| S_no   |
+-----+
| 202415121 |
| 202415122 |
+-----+
2 rows in set (0.027 sec)
```

4、查询姓名第二个字母是'e'的学生姓名

```
obclient(root@sys)[teaching_management]> select Name
-> from student
-> where Name like "_e%";
Empty set (0.035 sec)
```

5、查询全体男学生的学号、系、年龄结果按所在的系升序排列，同一系中的学生按年龄降序排列

```
obclient(root@sys)[teaching_management]> select s.No, d.Name, s.Age
-> from student as s, dept as d
-> where s.Gender = "Male"
-> and s.D_no = d.No
-> order by d.No asc, s.Age desc;
```

```
+-----+-----+-----+
| No      | Name          | Age  |
+-----+-----+-----+
| 202415121 | Computer Science | 20 |
| 202415120 | English        | 21 |
| 202415125 | English        | 19 |
+-----+-----+-----+
3 rows in set (0.114 sec)
```

```
obclient(root@sys)[teaching_management]> select student.No, dept.No, student.Age
-> from student, dept
-> where student.D_no = dept.No and student.Gender = "Male"
-> order by dept.No asc, student.Age desc;
```

```
+-----+-----+-----+
| No      | No  | Age |
+-----+-----+-----+
| 202415121 | 1  | 20 |
| 202415120 | 3  | 21 |
| 202415125 | 3  | 19 |
+-----+-----+-----+
3 rows in set (0.045 sec)
```

6、查询女学生的总人数和平均年龄

```
obclient(root@sys)[teaching_management]> select count(No), avg(Age) from student where Gender = "Female";
+-----+-----+
| count(No) | avg(Age) |
+-----+-----+
| 2         | 18.5000  |
+-----+-----+
1 row in set (0.018 sec)
```

7、查询选修 2 号课程并及格（分数大于 60）的学生的最高分、最低分及总分

```
obclient(root@sys)[teaching_management]> select max(Score), min(Score), sum(Score)
-> from score
-> where C_no = '2' and Score > 60;
+-----+-----+-----+
| max(Score) | min(Score) | sum(Score) |
+-----+-----+-----+
| 90         | 85         | 175        |
+-----+-----+-----+
1 row in set (0.012 sec)
```

8、向 scores 表中插入一条记录（200266666,1,85）

```
obclient(root@sys)[teaching_management]> insert into student
-> values(200266666,"Jas","Female",43,4);
Query OK, 1 row affected (0.022 sec)

obclient(root@sys)[teaching_management]> insert into score values (200266666,1,85);
Query OK, 1 row affected (0.022 sec)
```

9、求每个学生（号）的平均成绩，并将其超过 70 分的按学号输出

```
obclient(root@sys)[teaching_management]> select S_no, avg(Score)
-> from score
-> group by S_no
-> having avg(Score) > 70;
+-----+-----+
| S_no    | avg(Score) |
+-----+-----+
| 202415121 | 88.3333 |
| 202415122 | 85.0000 |
| 200266666 | 85.0000 |
+-----+-----+
3 rows in set (0.019 sec)
```

10、查询选修了课程 1 或者选修了课程 3 的学生姓名

```
obclient(root@sys)[teaching_management]> select distinct s.Name
-> from student as s, score
-> where s.No = score.S_no
-> and C_no in (1,3);
+-----+
| Name    |
+-----+
| Jas     |
| Tommy   |
| Michael |
+-----+
3 rows in set (0.034 sec)
```

11、查询既选修了课程 1 又选修了课程 2 的学生姓名

```
obclient(root@sys)[teaching_management]> (select distinct s.Name
-> from student as s, score
-> where s.No = score.S_no and C_no = 1)
-> intersect
-> (select distinct s.Name
-> from student as s, score
-> where s.No = score.S_no and C_no = 2);
+-----+
| Name |
+-----+
| Tommy |
+-----+
1 row in set (0.057 sec)
```

12、查询选修 Database 这门课最高分学生所在的系名

```
obclient(root@sys)[teaching_management]> select dept.Name from student, dept, course, score where score.C_no = course.No and
course.Name = "Database" and student.No = score.S_no and student.D_no = dept.No and score.Score >= all (select Score from
score, course where score.C_no = course.No and course.Name = "Database");
+-----+
| Name |
+-----+
| Computer Science |
+-----+
1 row in set (0.032 sec)
```

13、建立一个包含学生学号, 姓名, 年龄, 以及所在系名的视图 (赋予列名为 sno,sname,sage,deptname)

【create view】

```
obclient(root@sys)[teaching_management]> create view V as
-> select s.No as sno, s.Name as sname, s.Age as sage, d.Name as deptname
-> from student as s, dept as d
-> where s.D_no = d.No;
Query OK, 0 rows affected (0.476 sec)

obclient(root@sys)[teaching_management]> select * from V;
+-----+-----+-----+-----+
| sno      | sname  | sage | deptname      |
+-----+-----+-----+-----+
| 200266666 | Jas    | 43   | Management    |
| 202415120 | Michael | 21   | English       |
| 202415121 | Tommy  | 20   | Computer Science |
| 202415122 | Michael | 19   | Computer Science |
| 202415123 | Ale    | 18   | Mathematics    |
| 202415125 | Billy  | 19   | English       |
+-----+-----+-----+-----+
6 rows in set (0.019 sec)
```

出现的问题:

- 1.一开始定义的 dept 和 course 的 Name 属性字长（原本设定为了 varchar(11)）不够容纳附录中的数据（如"Computer Science"）
- 2.一开始直接往 score 里 insert 记录（200266666,1,85）时，由于 score 的 S_no 属性受外码完整性约束且 student 表里没有该学号，会报错

解决方案:

1. 用 alter table 重新修改了 dept 和 course 的 Name 属性，使之足以容纳数据里最长的名字
2. 先往 student 表中插入一个自定义的学号为 200266666 的学生，再往 score 里插入记录